Vehicle-Mounted Stepped-Frequency SAR — Design Outline

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Purpose and scope

Build a proof-of-concept, short-range, vehicle-mounted SAR using a USRP B210, a stepped-chirp waveform, two antennas, no RF front end, and an IMU for positioning. Success is a recognizably focused image of strong targets (corner reflectors, vehicles, building edges) from a short, slow drive. Perfection is not required.

- **In scope:** waveform design, B210 configuration, capture software, IMU time sync, data storage, offline image formation, validation.
- **Out of scope:** RF front-end hardware, real-time imaging, an X310 implementation (portability notes only).

How an agent should use this document:

- Values marked *start value* are tunable defaults.
- Values marked *measure* must be measured on the hardware before they are relied on.
- Items under Fixed constraints are decisions by the owner. Ask before changing them.
- Open questions in the Risks section need an owner answer before final configuration.

Fixed constraints

One SDR channel, two antennas and no RF hardware set the whole design: short range, a single receive channel, and calibration taken from antenna leakage rather than a reference loopback.

Item	Decision	Consequence for the design
SDR	USRP B210 (AD9361), USB 3, UHD installed	~56 MHz max instantaneous bandwidth per step; 70 MHz to 6 GHz tuning
Antennas	2: one TX (channel A TX/RX port), one RX (channel A RX2 port)	Quasi-monostatic geometry; direct TX-to-RX leakage is always present

Item	Decision	Consequence for the design
RF chain	None (no PA, LNA, filter, coupler)	Low TX power (~10 dBm), short range, no hardware reference channel
Waveform	Stepped chirp across several sub-bands	Wide total bandwidth; LO phase is random after every retune
Platform	Ground vehicle, short run	Slow, roughly straight, side-looking track
Navigation	High-grade IMU	Positions per pulse; requires tight time sync and a lever-arm model
Host	Laptop on the vehicle, USB storage	RAM buffering plus USB SSD is sufficient at the planned data rates
Quality bar	Proof of concept	Approximations (leakage calibration, autofocus) are acceptable

Only one receive channel is used, so the B210 can run at its full single-channel rate. The second channel stays free for a later reference loopback if hardware is added.

Key parameters and performance budget

The start values give about 0.5 m range resolution, a maximum speed near 0.7 m/s, and about 24 MB/s of data. All three trade against each other through the sweep time.

Parameter	Start value	How it is derived or used
Center frequency	~2.4 GHz (band TBD, see Risks)	$\lambda \approx 12.5$ cm; a lower band relaxes sampling and position accuracy
Steps per sweep (N)	6	More steps = more bandwidth, but slower sweeps
Step spacing	45 MHz	5 MHz overlap between adjacent 50 MHz chirps aids stitching
Chirp bandwidth per step	50 MHz	Fits inside the 56 MS/s sample rate
Total bandwidth (B)	~275–300 MHz	Range resolution $c / 2B \approx 0.5$ m

Parameter	Start value	How it is derived or used
Sample rate / master clock	56 MS/s	Single channel; at or below the B210 single-channel limit
Chirp length (T)	20 µs, repeated continuously	Unambiguous range cT/2 = 3 km, far beyond useful range
Dwell per step	1 ms (50 chirps)	Chirps averaged coherently for SNR
Retune + settle per step	5–10 ms (*measure*)	Dominates the sweep time
Sweep time	≈ N × (settle + dwell) ≈ 55 ms	~18 sweeps/s
Along-track sample spacing	⩽ λ/3 ≈ 4 cm	Per sub-band; avoids azimuth ambiguities
Max vehicle speed	spacing / sweep time ≈ 0.7 m/s	Drive at 0.3–0.5 m/s for margin
Position accuracy needed	≈ λ/8 ≈ 1.6 cm over the run	Residual error removed by autofocus
Data per sweep	N × dwell × 56 MS/s × 4 B ≈ 1.3 MB	sc16 IQ
Average data rate	≈ 24 MB/s	A 40 s run is about 1 GB
Azimuth resolution	≈ λR / 2L	~0.13 m at R = 20 m with a 10 m aperture

Recompute the sweep time, max speed and data rate whenever N, dwell, or the measured settle time changes. Keep these formulas in the configuration tooling so the agent can print the budget before every run.

RF and antennas without a front end

The main RF risk is TX-to-RX leakage saturating the receiver. The same leakage is the phase reference, so reduce it without eliminating it.

- **Range expectation:** ~10 dBm TX and modest antenna gain give useful returns from strong targets out to a few tens of meters. Long dwell and coherent averaging compensate partly.

- **Antenna mounting:**
  - Both antennas side-looking, same boresight, pointed slightly down toward the scene.
  - Side by side, separated as far as practical, never facing each other.
  - A metal plate or RF absorber between them if available.
  - Rigid mount: the TX-RX geometry must not flex during the run.
- **Gain setup:** start with low TX gain and moderate RX gain. Raise TX gain until the leakage peak is strong but the RX samples stay well below full scale (*measure*: peak magnitude < ~50% of full scale).
- **Fixed gains:** manual gain only, no AGC. Use the same gains for every step and every run.
- **DC and LO leakage:** the AD9361 is zero-IF. Tune with an LO offset (`tune_request_t(freq, lo_offset)`) so the chirp band avoids DC.
- **Static clutter:** reflections from the vehicle body sit at fixed ranges, like the leakage. They are removed in processing by subtracting the mean range profile across pulses.
- **Spurious emissions:** with no filter, harmonics and spurs radiate. Keep the sweep inside the permitted band and TX power as low as the range allows.

Waveform and timing

Transmit the same baseband chirp at every step, retune TX and RX together, wait for the LO to settle, then record a timestamped burst. Each burst becomes one pulse in processing.

Chirp

- Linear FM, 50 MHz span centered in baseband (offset from DC by the LO offset), 20 µs long, generated on the host as complex float or sc16.
- Apply a short amplitude taper (a few % raised-cosine) at the chirp edges to reduce spectral splatter.
- Store the exact transmitted baseband chirp in the dataset. Processing uses it as the matched-filter replica.

Per-sweep loop

1. For step k = 0…N−1: set TX and RX frequency to f_start + k × 45 MHz, with the same LO offset on both.
2. Wait for LO lock (poll the `lo_locked` sensor) plus a fixed settle margin (*measure*).

3. Issue a timed RX stream command (`STREAM_MODE_NUM_SAMPS_AND_DONE`) for dwell × sample rate samples.
4. Record the burst with its metadata: start `time_spec` , step index, TX/RX frequency, LO offset, gains, sweep index.
5. Repeat sweeps until the run ends.

TX behavior: simplest is to keep the TX streamer running continuously, looping the chirp buffer through retunes. Data recorded during settling is discarded anyway. An alternative is timed TX bursts aligned to the RX burst. Pick one, and record which in the metadata.

Timing details

- Set device time at start (`set_time_now` , or `set_time_next_pps` when a PPS is wired; see IMU section).
- Timed tune commands (`set_command_time`) reduce jitter if the B210 honors them for tuning. *Verify* on the hardware; otherwise retune from the host and rely on RX timestamps.
- Burst timestamps, not host clock readings, are the ground truth for pulse timing.
- Chirps within a dwell are periodic. Their start sample is known only up to the TX-to-RX pipeline latency, which the leakage calibration absorbs.

Phase calibration from antenna leakage

The AD9361 TX and RX synthesizers come up with a random relative phase after every retune. Without correction, sub-bands do not add coherently and the image will not focus. The direct antenna-to-antenna leakage, a fixed short path present in every burst, serves as the per-step reference.

Per burst:

1. Coherently average the chirps in the dwell into one pulse.
2. Range-compress it against the stored replica (FFT, multiply by the conjugate replica spectrum, IFFT).
3. Find the leakage peak in a small fixed window near zero range. Its location is found once from static data, then held fixed.
4. Divide the whole complex pulse by the leakage peak′s complex value (phase only, or phase and amplitude).

One-time correction: the leakage path is not exactly zero length. Normalizing to it shifts every step′s phase by $2\pi f_k \tau_{leak}$. Estimate τ_{leak} once:

- Fit the residual phase slope across steps on a corner reflector at a surveyed range, or
- Compute it from the measured antenna separation.

Then apply exp(+j2π f_k τ_leak) per step.

After calibration: subtract the mean calibrated pulse per step across the run. This removes leakage and vehicle-body returns before image formation.

Checks:

- Leakage amplitude should be stable within a few dB across steps and sweeps.
- Phase after normalization on a static scene should be repeatable to within a few degrees. *Measure* this on the bench first.

Upgrade option: a cabled TX-to-RX-B coupler or attenuator loopback would replace leakage as the reference. It costs the second channel and halves the sample rate.

Trajectory and the speed trade-off

Because the vehicle moves during a sweep, every sub-band burst is treated as its own pulse at its own position. The binding constraint is that each sub-band must be revisited within about λ/3 of travel.

No stop-and-go assumption. Do not stitch a sweep's sub-bands into one range profile at one position. Backproject each calibrated sub-band pulse using its own timestamp, IMU-derived TX and RX antenna positions, and center frequency. Coherent summation over all pulses produces the wide-band image.

Speed limit: v_max ≈ (λ/3) / sweep time.

Center freq	Spacing (≈ λ/3)	Sweep 55 ms	Sweep 100 ms	Sweep 200 ms
2.4 GHz	4.2 cm	0.76 m/s	0.42 m/s	0.21 m/s
5.8 GHz	1.7 cm	0.31 m/s	0.17 m/s	0.09 m/s

Levers when speed is too low:

- Fewer, wider steps.
- Shorter dwell.
- Lower center frequency.
- Shorter measured settle time.

- Accept mild undersampling. Ghosts fall at different azimuth positions for different sub-bands, so the coherent sum partly suppresses them and the image degrades gradually.

Driving guidance:

- Straight, steady path; constant speed preferred but not required.
- Scene broadside to the antennas.
- Run length sets the synthetic aperture: 10–20 m is enough for a first image.
- Start recording before the scene enters the beam and stop after it leaves.

Beamwidth: a wide beam gives a longer synthetic aperture and finer azimuth resolution, but tighter spacing requirements. The λ/3 rule assumes a wide (~60°+) beam. A narrower beam relaxes spacing to roughly $\lambda / (4 \sin(\theta/2))$.

IMU, time sync and lever arm

Every pulse needs TX and RX antenna positions accurate to about 1.6 cm relative to each other over the run. That requires a shared clock between the IMU and the B210, and a measured lever arm.

Time sync, best to worst:

1. **Shared PPS:** route the IMU/GNSS PPS output to the B210 PPS input. Call `set_time_next_pps` with the GNSS second so USRP timestamps and IMU time share an epoch. Target error < 0.1 ms.
2. **Host-clock bridge:** log the host monotonic clock against USRP time at start and periodically, and timestamp IMU messages on arrival. Expect around 1 ms error, which is about 0.5 mm at 0.5 m/s and acceptable here.
3. **Post-hoc alignment:** cross-correlate a deliberate motion event (start of travel, a bump) seen in both the IMU and the radar data. Fallback only.

Lever arm and attitude:

- Measure the 3D offset from the IMU reference point to each antenna's phase center (TX and RX separately) to ~5 mm.
- Position per antenna = IMU position + R(attitude) × lever arm. Pitch and roll on uneven ground matter: 1° over 0.5 m is ~9 mm.
- Treat the geometry as bistatic: two-way path = $|p_{tx} - x| + |p_{rx} - x|$. Do not collapse it to a single midpoint unless the antennas are within a few cm.

Logging:

- Record raw and navigation-solution IMU output at full rate (≥ 100 Hz; 200+ Hz preferred), with time-quality flags.
- Export position in a local ENU or along-track frame for processing.
- Interpolate (cubic or spline) positions and attitude to each burst's center time.

Accuracy expectation: a high-grade IMU over 10–60 s is likely at the cm level. GNSS/RTK aiding improves absolute accuracy, but relative smoothness along the track matters most. Autofocus in processing is expected to remove the residual.

Capture software architecture

Use one capture program on the UHD `multi_usrp` API. C++ is preferred; a Python prototype is acceptable at these duty cycles. It has four threads, writes to a preallocated RAM buffer during the run, and flushes to disk afterward.

Threads:

1. **Control:** runs the sweep loop (retune, lock wait, timed RX command), owns the configuration, and prints the parameter budget at start.
2. **TX streamer:** loops the chirp buffer continuously. Logs underflows (`U`) with timestamps.
3. **RX streamer:** calls `recv()` into slots of the RAM buffer, one slot per burst, plus a metadata record. Checks `rx_metadata_t` on every call.
4. **IMU logger:** reads the IMU interface and writes timestamped records into its own buffer.

Error handling:

- Overflow (`O`), sequence error (`D`), timeout, or late command → mark that burst invalid in metadata. Never silently drop or shift it.
- Report per-run counts of invalid bursts. Above ~1%, treat the run as suspect.
- LO not locked within the timeout → mark the step invalid and continue.

Configuration: a single YAML or JSON file holds all of the following, and the program copies it into the dataset:

- Device args (`num_recv_frames` , `recv_frame_size`) and master clock.
- Rates, frequencies, steps, LO offset, gains, chirp parameters, dwell.
- Sync mode, IMU port, lever arms.

Host setup:

- B210 on its own USB 3 controller, no hub; USB autosuspend disabled.
- CPU governor `performance`; `uhd::set_thread_priority_safe()` on the streamer threads.
- Laptop on AC power or with power management disabled.

Portability: keep device-specific values in config only so the same program runs on an X310 by changing device args and rates.

Data storage over USB and file format

USB storage is sufficient. At about 24 MB/s average, a whole run fits in RAM, and the USB SSD only needs to absorb a write after the run.

Strategy:

- Preallocate (and `mlock`) a RAM buffer sized for the maximum run length plus 25%. A 60 s run is about 1.5 GB.
- Flush to a USB 3 SSD after the run ends, using large sequential writes.
- Streaming to disk during the run is optional. If used, put the SSD on a different USB controller from the B210, since sharing a controller is the most common cause of overflows.
- Continuous capture at the full 56 MS/s (~224 MB/s) is also within a USB 3 SSD's ability, but not needed for this waveform.

Dataset layout, one folder per run:

File	Contents
`iq.sc16`	Raw RX bursts back to back, interleaved int16 I/Q
`bursts.csv` (or Parquet)	Per burst: index, sweep, step, start sample offset in file, USRP timestamp, TX/RX frequency, LO offset, gains, valid flag, error codes
`iq.sigmf-meta`	SigMF metadata: sample rate, datatype, hardware, global parameters
`chirp_replica.cf32`	The exact transmitted baseband chirp

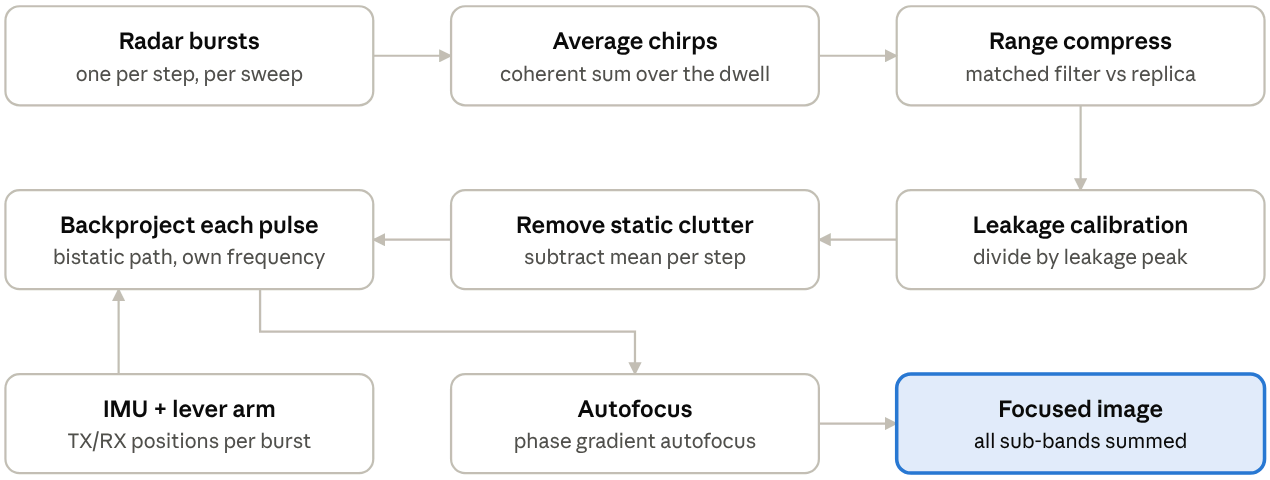
File	Contents
`imu.csv`	Timestamped IMU navigation solution (and raw data if available)
`config.yaml`	The full run configuration, as used
`run_log.txt`	UHD messages, error counts, sync status, budget printout

Integrity: write a checksum per file and a run summary: burst counts, invalid percentage, duration, distance traveled.

Processing pipeline

Processing is offline, in Python (NumPy, with an optional GPU path via CuPy). Each sub-band burst is calibrated independently, then backprojected at its own position and frequency. The bands combine only in the image.

Each sub-band burst is calibrated, then backprojected at its own position



processing chain · 8 stages, IMU side input

IMU positions enter at backprojection; everything before it works on one burst at a time.

1. **Load and filter:** read bursts and metadata; drop invalid bursts; interpolate IMU positions and attitude to each burst's center time.
2. **Average chirps:** coherently sum the chirps within each dwell into one pulse per burst.

3. **Range compress:** matched filter against the stored replica, with a window (e.g. Hamming) for sidelobe control. Upsample range profiles 4–8× for interpolation accuracy.
4. **Leakage calibration:** per-burst normalization plus the one-time τ_{leak} correction (see the calibration section).
5. **Static clutter removal:** subtract the mean calibrated profile per step across the run.
6. **Backprojection:** for each pixel x on a ground-plane grid and each pulse i at center frequency f_i, add $s_i(R) \cdot \exp(+j\,2\pi\, f_i\, R / c)$, where $R = |p_{tx,i} - x| + |p_{rx,i} - x|$ and s_i is interpolated at R. Grid spacing should be finer than half the range and azimuth resolution.
7. **Autofocus:** phase gradient autofocus, or a sharpness-maximizing per-pulse phase correction, to remove residual position and timing error.
8. **Output:** magnitude image in dB with geo-referenced axes, plus a per-sub-band image set for debugging calibration.

Test and validation plan

Validate in six stages, from bench to scene. Do not move to the next stage until the current one passes.

Stage	Test	Pass criterion
1. Bench timing	Run sweeps with the vehicle parked; measure retune and settle time, overflow rate	Settle time known; < 0.1% invalid bursts over 5 min
2. Leakage calibration	Static scene, many sweeps; apply per-step normalization	Calibrated phase repeatable to a few degrees; leakage stable within a few dB
3. Static range	Corner reflector at 2–3 surveyed ranges, vehicle parked; stitch sub-bands	Peak at the correct range within one resolution cell; mainlobe ≈ 0.5 m; τ_{leak} estimated
4. Sync and motion	Drive past nothing in particular; compare IMU motion start with radar static-clutter change	IMU and radar time agree to ≤ 1 ms

Stage	Test	Pass criterion
5. Point target drive	Slow drive (0.3 m/s) past one isolated corner reflector at 10–20 m	Focused point after backprojection and autofocus; azimuth width near the predicted value
6. Scene	Drive past a real scene (cars, building edges)	Recognizable structure

Risks, open questions and upgrade path

The largest unknowns are the legal operating band and the real B210 retune time. Both change the parameter budget and need answers before the final configuration.

Risk	Effect	Mitigation
Retune and settle slower than assumed	Longer sweeps, lower max speed	Measure first; reduce steps; drive slower
Leakage saturates RX	Clipped data, no usable calibration	Lower TX gain; separate and shield the antennas
Leakage too weak or unstable	Poor per-step phase calibration	Raise TX gain slightly; fall back to a cabled reference on channel B
AD9361 internal calibrations change phase or gain between steps	Residual sub-band errors	Leakage normalization absorbs most of it; keep gains fixed
USB overflows	Gaps in the aperture	RAM capture; separate controllers; mark invalid bursts
IMU/radar time offset	Defocus, smeared azimuth	PPS sync; motion-event cross-check; autofocus
Unmodeled lever arm or attitude	Defocus that grows with vehicle motion	Measure the lever arm; apply attitude per pulse
Ground multipath and clutter	Ghosts, reduced contrast	Tilt the antennas; accept for proof of concept

Open questions for the owner:

- [ ] Operating band and legal basis: US ISM 2.4 GHz is only ~83.5 MHz wide; amateur bands need a license. This sets the center frequency, steps and total bandwidth.
- [ ] Antenna type and beamwidth (sets the azimuth spacing rule).
- [ ] IMU model, output interface, and whether it has a PPS output and GNSS aiding.
- [ ] Expected target ranges and scene type.

X310 upgrade path:

- 10 GbE and up to ~160 MHz per step with UBX-160 daughterboards. That means far fewer steps for the same bandwidth, so faster sweeps and higher allowed speed.
- Phase-synchronized daughterboard options may remove the need for per-step leakage calibration. Keep the calibration code as a fallback.
- Higher data rates (hundreds of MB/s per channel): keep the RAM-first capture design, and use an internal NVMe drive for longer runs.
- Only config should change (device args, rates, steps); the capture and processing code stays the same.

Agent task breakdown

Work in this order. Each task ends with a runnable deliverable and a check against the validation plan.

- [ ] **T1 Config and budget tool:** a schema for the run config plus a script that prints the derived budget (resolution, sweep time, max speed, data rate) and warns on violations.
- [ ] **T2 Chirp generator:** produce the tapered baseband chirp and its replica file from the config.
- [ ] **T3 Bench timing probe:** measure retune, lock and settle times and overflow behavior on the B210; write the results back into the config. Validation stage 1.
- [ ] **T4 Capture program:** control, TX, RX and IMU threads; RAM buffer; dataset writer in the layout above; invalid-burst marking.
- [ ] **T5 IMU interface and sync:** IMU reader, PPS or host-clock sync, logger, and an interpolation utility that returns per-burst TX/RX antenna positions using the lever arms.

- [ ] **T6 Range compression and leakage calibration:** chirp averaging, matched filter, per-step leakage normalization, τ_leak estimation, static-clutter removal. Validation stages 2–3.
- [ ] **T7 Backprojection imager:** per-pulse, per-sub-band bistatic backprojection on a ground grid, vectorized NumPy with an optional GPU path. Validation stage 5.
- [ ] **T8 Autofocus:** phase gradient autofocus (or a simple image-sharpness optimizer) applied after backprojection.
- [ ] **T9 Simulator:** synthetic point targets along an IMU-like trajectory with random per-step LO phases, so T6–T8 can be tested before field data exists.
- [ ] **T10 Field runbook:** pre-run checklist (gains, sync, lever arms, budget printout), run procedure, post-run integrity check.

Build T9 early. It lets the processing chain be developed and verified independently of hardware access.